DRAFT STANDARD OPERATING PROCEDURE

Sugarcane Germplasm Field Layout and Conservation

Sugarcane Breeding and Germplasm Conservation Program

SOP No.	SCB-GC-SOP-04	**Revision**	0
Effective date	_____	**Supersedes**	New
Prepared by	_____	**Reviewed by**	_____
Approved by	_____	**Controlled copy**	Yes / No

Status: Draft technical SOP. Field dimensions, thresholds, chemical treatments, quarantine steps, and local compliance requirements must be validated and approved by the implementing institution before adoption.

1. Purpose

To conserve sugarcane genetic resources in a field genebank while maintaining accession identity, plant health, genetic integrity, accessibility for characterization and breeding, and recoverability through safety duplication.

2. Scope

Applies to clonally maintained Saccharum species, commercial hybrids, pre-breeding/introgression lines, allied genera, genetic stocks, and other vegetatively conserved sugarcane materials.

3. Conservation principles

- Identity before quantity: a large collection with uncertain labels has low breeding value.
- Redundancy for irreplaceable material: avoid maintaining a unique accession at only one location when feasible.
- Health surveillance is continuous; germplasm fields must not become reservoirs of major pests and diseases.
- Field records, physical labels, and digital database must remain synchronized.
- Regenerate before vigor or stand quality is seriously compromised.

4. Functional collection structure

Arrange accessions into logical sections to simplify inventory and specialized management. A practical structure is Saccharum officinarum; S. spontaneum; S. robustum; S. sinense/S. barberi; other Saccharum species; local commercial/breeding hybrids; foreign hybrids; introgression/genetic stocks; allied genera; and newly introduced material after clearance.

5. Site and layout requirements

- Well-drained or appropriately drained site matched to local cane agronomy; irrigation capability where required.
- Secure access and clear perimeter so accidental harvesting or mixing is minimized.
- Permanent block/row markers and a surveyed or GPS-referenced field map.
- Access alleys for inspection, roguing, photography, and seed-cane cutting.
- Separate holding/quarantine or observation area for material not yet fully integrated into the active collection.

6. Plot configuration

A two-row accession plot is a practical starting configuration where land permits. Row length may commonly be set around 3-6 m for active conservation/characterization, but the institution should validate dimensions based on local vigor, land availability, and the need to maintain enough stalks for regeneration. Highly rare or weak accessions may require customized plot size or additional backup.

7. Field coding system

Use a location code independent of the accession number, for example GC26-B03-R12-P05. The permanent accession ID identifies the germplasm; the field code identifies only its current physical location. Never replace the permanent accession number when the material moves to another field.

8. Establishment procedure

1. Prepare the final field map and plot list before seed cane is cut.
2. Verify seed-cane source and identity using a two-person check for high-value or irreplaceable accessions.
3. Use clean cutting tools and plant-health sanitation procedures appropriate to local disease risks.
4. Keep each accession physically separated and labeled from cutting through transport, planting, and mapping.
5. Install a durable accession label and field-coordinate label immediately at planting; temporary labels may supplement but never replace the permanent marker.
6. Record planting date, source field, quantity planted, responsible personnel, and any identity or health concerns.
7. Conduct an establishment audit after emergence and correct map/label discrepancies before routine maintenance begins.

9. Routine monitoring

- Survival and missing stools.
- Off-types, mixtures, or unexpected phenotype.

- Smut and other locally important fungal diseases.
- Mosaic/viral symptoms and other systemic disorders.
- Borers, scales, and other insects relevant to the site.
- Lodging, flowering, ratoon decline, and abnormal growth.
- Label condition and map accuracy.

10. Off-type and contamination management

Do not immediately destroy a suspected off-type. Tag it, photograph it, compare the expected descriptor profile, inspect neighboring plots for possible mechanical mixing, and review passport/planting records. Where justified, use molecular identity testing. Remove only after the curator determines that the plant is contamination or an unintended mixture; document the decision.

11. Regeneration and movement

Regenerate accessions before stand decline threatens loss. Prepare destination maps in advance, cut only verified true-to-type stools, and use a two-person label/identity check at harvest and planting. Every movement must generate a source-to-destination record. Maintain previous field locations in the accession history rather than overwriting them.

12. Safety duplication

Priority and irreplaceable accessions should, where resources permit, have a duplicate in a second field location or another conservation system such as in-vitro culture. ICAR-SBI reports maintaining its world sugarcane collection as a field genebank while also conserving vulnerable S. officinarum accessions in vitro, illustrating a useful layered conservation approach.

13. Quality-control checkpoints

- 100% of active plots have readable accession and location labels.
- Field map matches the physical field after every regeneration or major casualty event.
- Suspect identity is flagged in the database until resolved.
- High-value accessions have documented safety-duplicate status where feasible.
- Regeneration retains source field, date, operator, and destination traceability.
- Annual inventory reconciles active, missing, dead, quarantined, duplicated, and lost accessions.

14. Annual field-genebank inventory sheet

Accession	Location	Stand %	Health status	Off-type?	Flowering	Regenerate ?	Safety duplicate	Remarks

Accession	Location	Stand %	Health status	Off-type?	Flowering	Regenerate ?	Safety duplicate	Remarks

References and technical basis

ICAR-Sugarcane Breeding Institute Research Centre, Kannur - world collection and in-vitro backup - Web source

FAO/Bioversity Multi-Crop Passport Descriptors v2.1 - conservation/storage data standard - Web source